

THE $Z \rightarrow \mu\mu$ CROSS SECTION WITH CMS 2016 OPEN DATA

SingleMuon, Run2016G+H, 16.39 fb^{-1} at 13 TeV – methods, results and validation – 15 September 2026

Result (dressed muons, $p_T > 26/20 \text{ GeV}$, $|\eta| < 2.4$, $60 < m_{\mu\mu} < 120 \text{ GeV}$):

$$\sigma_{\text{fid}}(pp \rightarrow Z/\gamma^* \rightarrow \mu\mu) = 790.1 \pm 0.2 (\text{stat}) \pm 8.5 (\text{syst}) \pm 9.6 (\text{lumi}) \text{ pb}$$

$\mu_Z = 0.988 \pm 0.016$ relative to aMC@NLO (799.6 pb); $\sigma(60 < m < 120) = 1931 \pm 33 \text{ pb}$; $\sigma(m > 50) = 2002 \pm 34 \text{ pb}$.

How

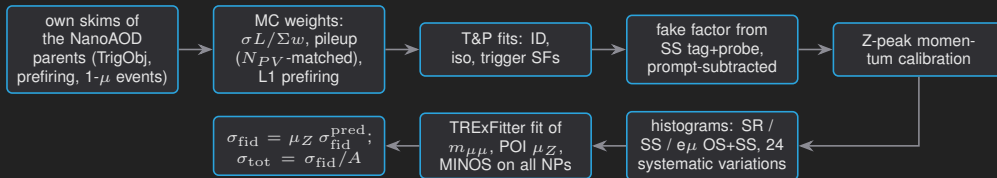
- own skims of the NanoAOD parents; simulation for signal and prompt backgrounds, no QCD MC
- corrections measured in place: pileup, L1 prefiring, ID / isolation / trigger scale factors from tag-and-probe fits, muon momentum from the Z peak, non-prompt muons from a data-driven fake factor
- TRExFitter v1.8.0 profile-likelihood fit of $m_{\mu\mu}$ ($12 \times 5 \text{ GeV}$ bins, 24 nuisance parameters), cross-checked by counting and by refits with other binnings

Sections

1. Strategy and inputs
2. Result and uncertainties
3. Validation
4. Conclusions

Code and numbers: `z-mumu/` (README.md, docs/09-15, output/v2/RESULTS_v2.md); re-view and fixes: REVIEW.md

1. STRATEGY AND INPUTS

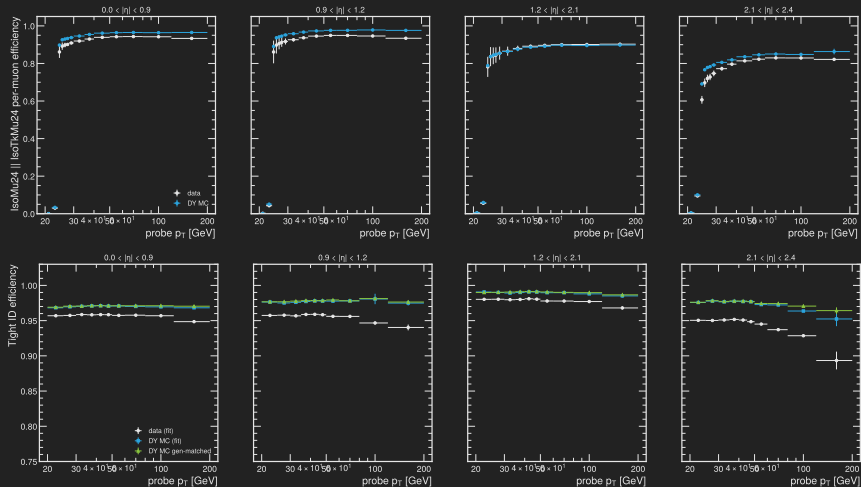


Selection (signal region)

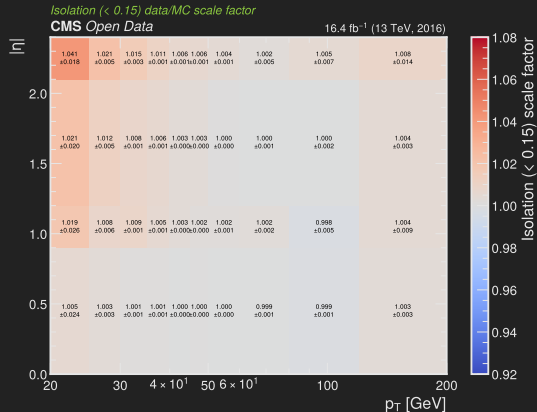
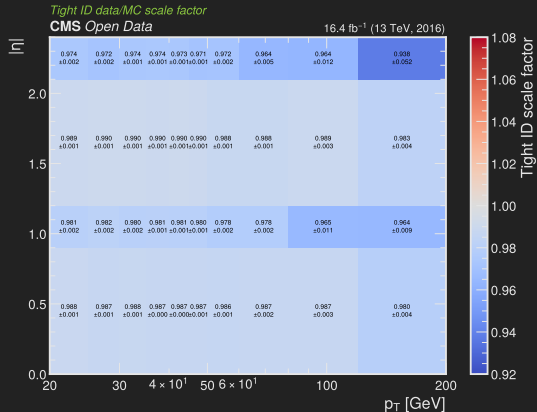
- HLT_IsoMu24 || HLT_IsoTkMu24, MET filters, $N_{PV} \geq 1$
- exactly two tight muons (tightId, iso < 0.15, $|d_{xy}| < 0.2$, $|d_z| < 0.5$ cm), $p_T > 26/20$, $|\eta| < 2.4$, ≥ 1 trigger-matched, opposite sign
- FSR-recovered $60 < m_{\mu\mu} < 120$ GeV: 10 378 567 data events, 99.3 % signal

Fiducial volume and normalisation

- dressed muons ($\Delta R < 0.1$), $p_T > 26/20$, $|\eta| < 2.4$, $60 < m < 120$ GeV; $C = 0.7914$, $A_{60-120} = 0.4092$ (aMC@NLO)
- $\sigma(Z/\gamma^* \rightarrow \ell\ell, m > 50) = 6077.22/3$ pb, $L = 16393.381$ pb⁻¹ (normtag)
- backgrounds: $Z \rightarrow \tau\tau$, $t\bar{t}$, tW , WW , WZ , ZZ from MC (0.62 %), non-prompt from data (0.04 %)



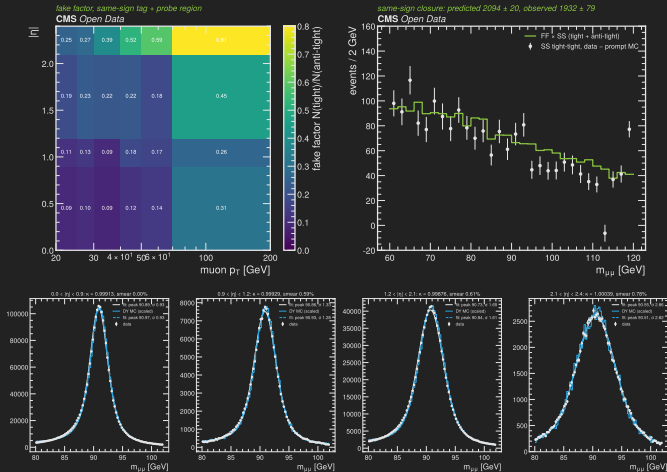
20.1 M tag-probe pairs. Trigger: trigger-object matching, per-muon plateau 0.907 (data) / 0.923 (MC), applied per event as $1 - (1 - \epsilon_1)(1 - \epsilon_2)$. Tight ID: simultaneous pass/fail fits (DY template $\otimes$ Gaussian + exponential); the MC fit closes on the gen-matched truth to -0.1% .



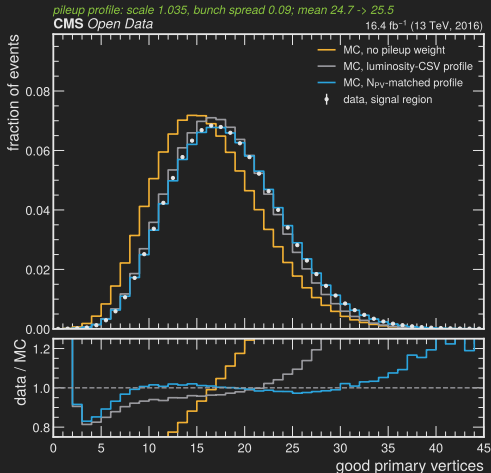
tight ID: 0.97–0.99, $\langle SF \rangle = 0.980$

isolation: 1.00–1.04, $\langle SF \rangle = 1.006$, flat in pileup (half-spread 0.003)

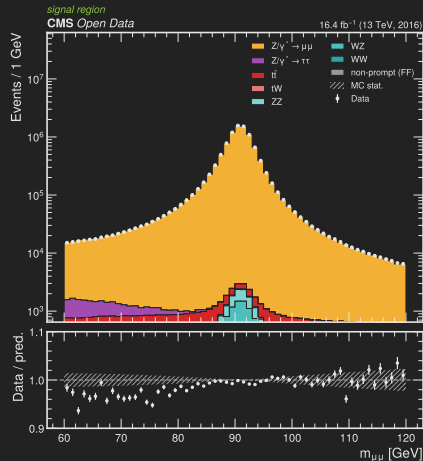
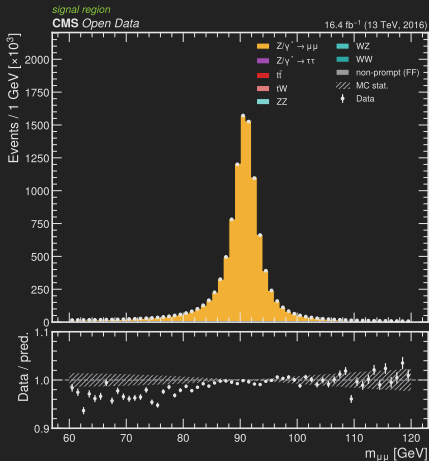
Systematic variants (background model, fit range, tag selection, powhcg templates) enter as the MuonID / MuonIso / MuonTrigger nuisance parameters, coherent over all cells. Muon reconstruction (track $\rightarrow$ muon) is not measurable in NanoAOD: external 0.4 % per muon, correlated, i.e. 0.8 % per event.



Fake factor $N_{\text{tight}}/N_{\text{anti-tight}}$ from same-sign tag+probe pairs with prompt MC subtracted: SR non-prompt 3870 ± 80 (stat) $\pm 30\%$ (method); same-sign closure 2094 ± 20 predicted vs 1932 ± 79 observed. Momentum: scale κ and extra smearing per $|\eta|$ from Breit-Wigner $\otimes$ Gaussian fits of the Z peak (data vs MC); propagated as $\text{MuonScale} / \text{MuonRes}$.

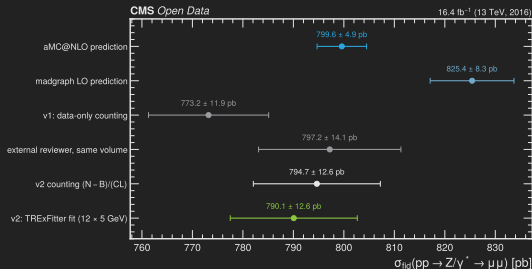


- No official `puWeights` file is reachable; the data profile comes from the per-luminisection `avgpu` of the Open Data luminosity record ($\times 69.2/80$ mb).
- That raw profile has no bunch-to-bunch spread and is $\approx 4\%$ too low: data/MC in N_{PV} rose from 0.8 to 1.2 (grey).
- Two parameters – a scale of the per-luminisection mean and a relative Gaussian spread – are fitted to the signal-region N_{PV} distribution: scale 1.035, spread 9%; χ^2/ndf 104 414 $\rightarrow$ 4 729 / 50 (blue).
- The Pileup nuisance parameter ($\sigma_{\text{mb}} \pm 4.6\%$) is then pulled by $+0.06\sigma$ instead of -1.5σ ; impact on σ_{fid} 0.13%.



10378567 data events; prompt MC + fake factor 10442328 pre-fit ($\mu_Z = 1$). The 3% deficit at 60–80 GeV is a lineshape difference between the data and aMC@NLO (section 2).

2. RESULT AND UNCERTAINTIES

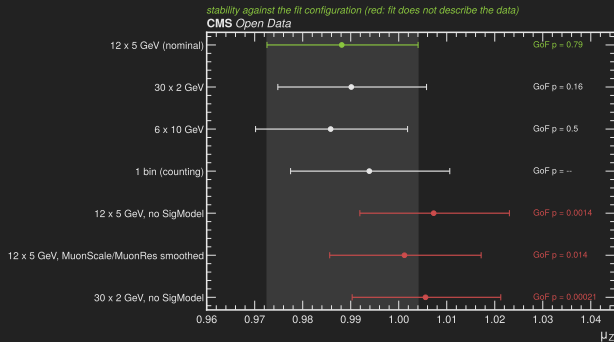


quantity	value
σ_{fid} (fit, 12×5 GeV)	790.1 ± 12.6 pb
σ_{fid} (counting $(N - B)/(CL)$)	794.7 pb
μ_Z	$0.988^{+0.016}_{-0.016}$
goodness of fit (12 bins)	$p = 0.79$
$\sigma(60 < m < 120), A = 0.4092$	1931 ± 33 pb
$\sigma(m > 50), A = 0.3947$	2002 ± 34 pb
C (reco / fiducial)	0.7914
aMC@NLO $\sigma_{\text{fid}}^{\text{pred}}$	799.6 pb
external reviewer, same volume	797.2 pb

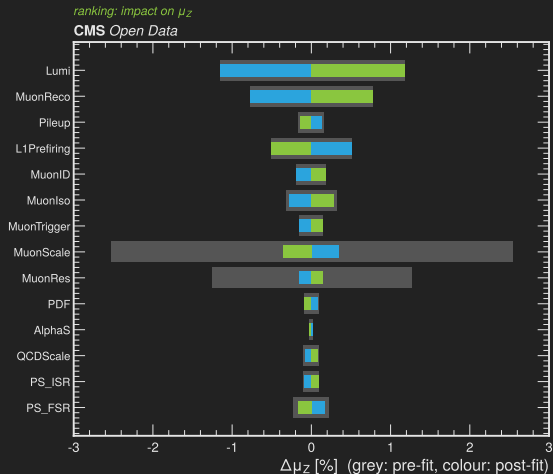
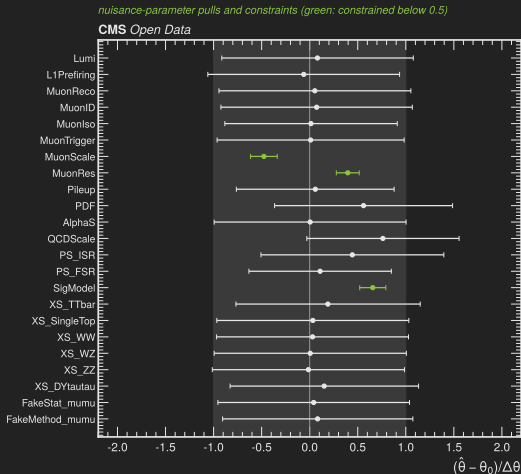
- $\sigma_{\text{fid}} = \mu_Z \times \sigma_{\text{fid}}^{\text{pred}}$; algebraically $(N - B)/(LC)$, so the assumed DY cross section cancels.
- Fit and counting differ by 0.6%: the shape information moves the lineshape and momentum nuisance parameters (next slides).
- The data statistical uncertainty is 0.03%; the effective statistical limit is the MC (0.38%).

group	on μ_Z	corr.
luminosity (external)	1.20 %	yes
muon efficiency (reco 0.77 %)	0.87 %	μ ch.
L1 prefiring	0.51 %	yes
signal modelling	0.49 %	yes
MC statistics	0.38 %	no
muon momentum	0.37 %	μ ch.
pileup	0.13 %	yes
background normalisation	0.08 %	yes
non-prompt	0.04 %	no
data statistics	0.03 %	no
total	1.57 %	

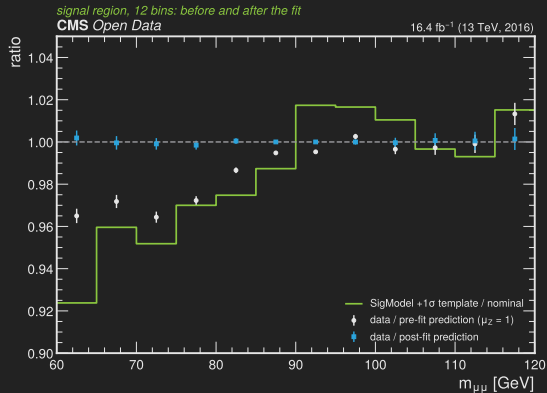
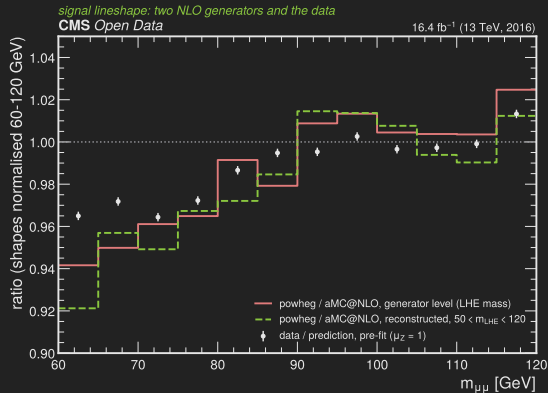
1.06 % without the luminosity. Acceptance for σ_{tot} : 0.61 % (PDF, scales, α_s).



The binnings that describe the data agree within ± 0.2 %. Dropping the generator template or smoothing the momentum templates gives fits with $p \leq 0.01$: rejected.



MINOS on every parameter. Only the three shape parameters are constrained by the 10 M-event spectrum: $\text{SigModel} +0.66$ (0.14), $\text{MuonScale} -0.48$ (0.14), $\text{MuonRes} +0.40$ (0.12); everything else within $\pm 0.8\sigma$ at its prior width. Luminosity and muon reconstruction dominate the ranking.

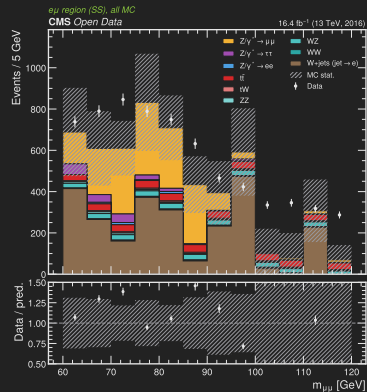
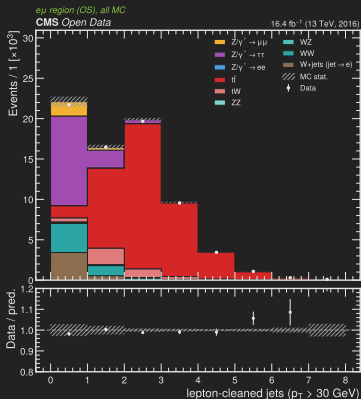
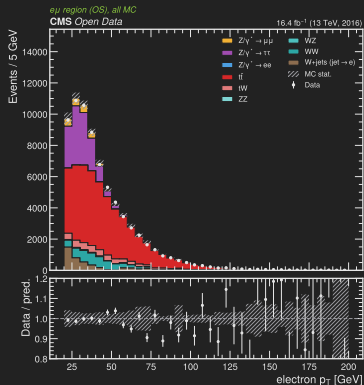


Left: powheg has 4–6% less continuum than aMC@NLO below 80 GeV already at Born level; the data sit between the two. Right: the two-sided SigModel template (powheg / aMC@NLO inside the common $50 < m_{LHE} < 120$ GeV window) lets the fit absorb it with a $+0.66\sigma$ pull; post-fit the spectrum is described within 0.3% in every bin.

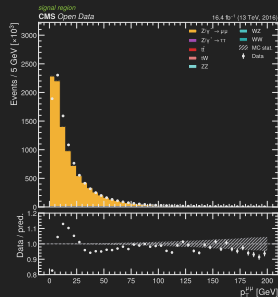
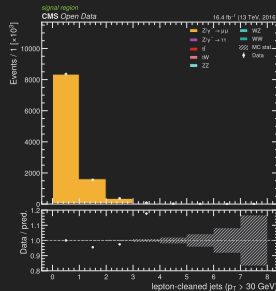
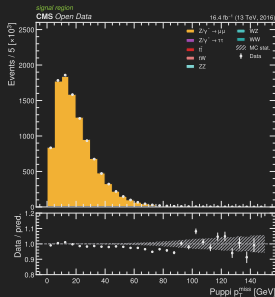
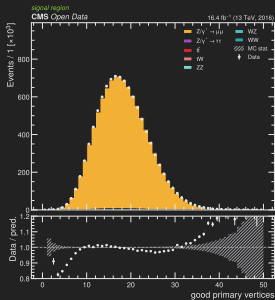
3. VALIDATION

E_{μ} REGIONS: FLAVOUR-SYMMETRIC BACKGROUNDS AND NON-PROMPT ELECTRONS

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One tight matched muon + one medium electron, $60 < m_{e\mu} < 120$ GeV. Opposite sign: 72 357 data vs 72 959 predicted (0.992), $t\bar{t}$ 43 165, $Z \rightarrow \tau\tau$ 13 914, WW 5 274, tW 4 113, non-prompt electrons (W +jets, $Z \rightarrow \mu\mu$ +conversion) ≈ 6 000. Same sign (right): 6 718 vs 5 428, the charge-symmetric non-prompt part, limited by W +jets MC statistics and the absent QCD sample. Validation only (not in the fit).



Pileup after the matched profile; Puppi p_T^{miss} (PF p_T^{miss} is broader in data: MC jets are not JER-smeared and the unclustered energy is mismodelled – neither is used); lepton-cleaned jets; $p_T^{\mu\mu}$ shows the known aMC@NLO+Pythia8 low- p_T mismodelling, which enters C through the p_T thresholds and is covered by the generator comparison.

WHAT THE REVIEW CHANGED (SAME DAY)

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finding	fix	effect on the result
$e\mu$ region +9 %: non-prompt electrons removed by the prompt-only MC filter	keep all MC in the $e\mu$ regions, add W+jets and a same-sign $e\mu$ region	none (validation)
30×2 GeV fit not robust (± 0.8 % with binning / templates), over-constrained NPs, one-sided SigModel with a generator-cut artefact	12×5 GeV, no smoothing, two-sided SigModel inside the common LHE window, MINOS on all parameters, stability table	$791.8 \rightarrow 790.1$ pb; stable to ± 0.2 %
pileup profile 4 % low, Pileup pulled -1.5σ	profile fitted to N_{PV}	pull $+0.06\sigma$
MuonReco 0.4 % per event instead of per muon	0.8 % per event	syst $1.40 \rightarrow 1.57$ %
jets not lepton-cleaned; p_T^{miss} broader in data	cleaned jets; Puppi p_T^{miss} shown, nothing else needed	none

Full account with reasoning: `z-mumu/REVIEW.md` (section 0 = status after the fixes).

- $\sigma_{\text{fid}}(pp \rightarrow Z/\gamma^* \rightarrow \mu\mu) = 790.1 \pm 0.2 \pm 8.5 \pm 9.6 \text{ pb}$, 1.6 % total, in agreement with aMC@NLO (799.6 pb, $\mu_Z = 0.988 \pm 0.016$) and with an independent analysis of the same data (797.2 pb).
- Every correction is measured in the data itself (efficiencies, momentum scale, pileup, non-prompt background); the only external inputs are the luminosity (1.2 %) and the muon reconstruction efficiency (0.4 %/muon) – together 1.4 of the 1.6 %.
- The profile-likelihood fit is validated three ways: goodness of fit $p = 0.79$, stability of μ_Z against the binning ($\pm 0.2 \%$), and the counting cross-check (794.7 pb).
- The $e\mu$ regions validate the flavour-symmetric backgrounds to 1 %; the same-sign regions validate the non-prompt estimate.
- Inputs for the combination: `fit/fitinputs/zmumu.root`, `fit/zmumu.config` (POI `mu_Z`, correlated NP names per `fitting/CONVENTIONS.md`); $A_{60-120} = 0.4092$ recommended.

Open items

- the 60–80 GeV lineshape: data between aMC@NLO and powheg; a third generator or NLO EW corrections would settle it
- official pileup, prefiring, Muon-POG and Rochester inputs need CERN credentials; the in-house versions are documented stop-gaps
- no electron scale factors and a 24 % under-prediction of the same-sign $e\mu$ region (W+jets MC statistics, no QCD sample); validation only
- the acceptance convention (60–120 vs $m > 50$; NLO vs LO, 3.2 %) is a decision for the combination